

## src/components/KnowledgeGraphPanel.tsx

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### Overview

A new `KnowledgeGraphPanel` component is added at `src/components/KnowledgeGraphPanel.tsx` (lines 1-692). It provides a UI for building, viewing, and interacting with a knowledge graph.

### Key changes

- **Imports & lazy loading** – Lines 1-17 add `createPortal`, `FileChange`, `LlmSettings`, `DocWorkspaceMetrics`, `isNovadiffDocsReservedPath`, and lazily import `NovaDiffGraphExplorer` (lines 22-24).
- **Props interface** – Defined at lines 26-41 ( `KnowledgeGraphPanelProps` ) with fields for comparison flags, root paths, titles, file changes, LLM settings, metrics, and UI flags.
- **Utility functions** – `parseProgressMessage` (lines 62-68) extracts `current / total` from a string; `buildAutoKey` (lines 70-76) creates a memoized key for caching.
- **Error boundary** – `GraphExplorerErrorBoundary` (lines 79-115) catches explorer render errors and offers a retry button that resets the boundary.
- **State & effects** – Lines 133-147 hold state for build side, progress, stats, payload, viewer, fullscreen, and activity. `useEffect` hooks (lines 176-206, 305-334, 342-358, 380-393) manage progress listening, build lifecycle, fullscreen handling, and mounting.
- **Build & load logic** – `runBuild` (lines 227-303) triggers graph construction, uses `loadGraphFromDisk` (lines 208-225) for caching, and updates `buildGenerationRef` to avoid stale updates.
- **Viewer & portal** – `viewerShell` (lines 469-532) renders the explorer; `createPortal` is used for fullscreen mode (lines 676-679).
- **UI elements** – Toolbar buttons, progress card, stats grid, and status line are rendered conditionally based on state.

### Impact

- The component introduces many hooks and state variables, increasing render complexity.
- `buildGenerationRef` mitigates race conditions by ensuring only the latest build updates the UI.
- Lazy loading and `requestIdleCallback` (lines 328-332) defer heavy work until idle.
- Background activity integration ( `useBackgroundActivityOptional` , line 150) reports progress to the UI.

### Risks & follow-ups

- Verify that `@novadiff/graph-view` resolves in all target environments.
- Ensure the `requestIdleCallback` fallback timer behaves correctly on browsers lacking the API.

- Confirm that `createPortal` does not interfere with server-side rendering or test setups.
- Test that the error boundary's retry button correctly resets the explorer state.
- Validate that all `window.electronAPI` calls handle missing or malformed responses gracefully.